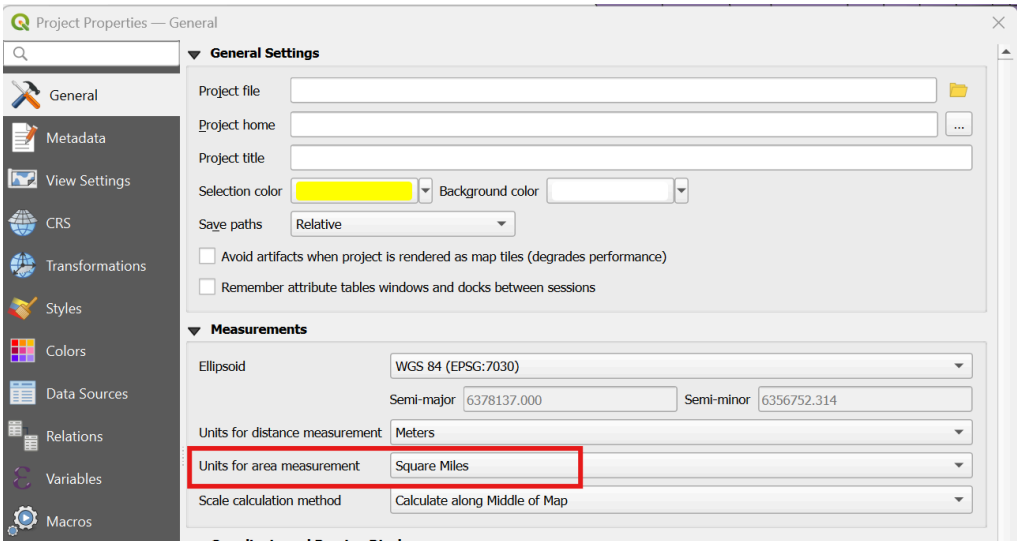
Use QGIS Field Calculator: Calculate Polygon Area

- Open **QGIS Desktop** and add the vector layer.
- In the Layers Panel, right-click on the layer and select **Open Attribute Table**.
- In the attribute table, click on the **Toggle editing mode** button  . The editing mode allows you to make changes to the attribute table, such as *creating a new column* or *deleting an existing column.*
- Now, we are about to create a new field to calculate polygon areas in **square miles**. Before doing this in QGIS, we need to figure out the area unit.
  - In the main QGIS window, select **Project > Properties > General**.
  - The default area measurement unit is shown under **Measurements.** The Chicago layer is in Square Meters. Use the dropdown menu to select **Square Miles** and click **OK** to apply the change.

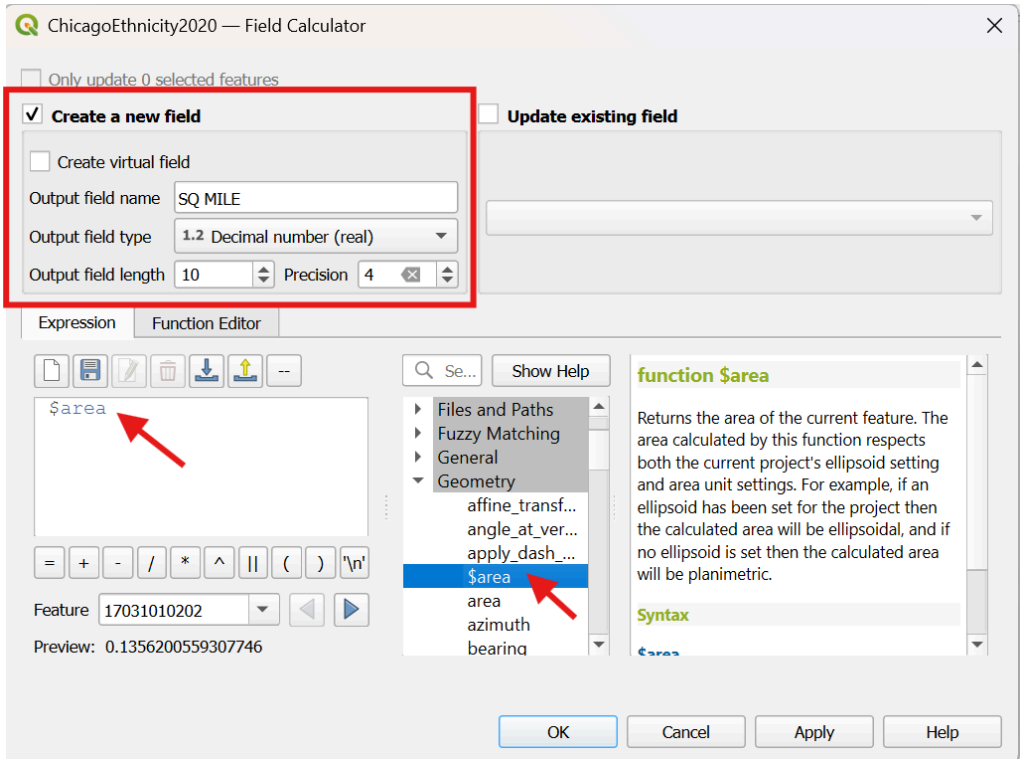


- Return to the layer's attribute table, click on the **Open field calculator** button  .

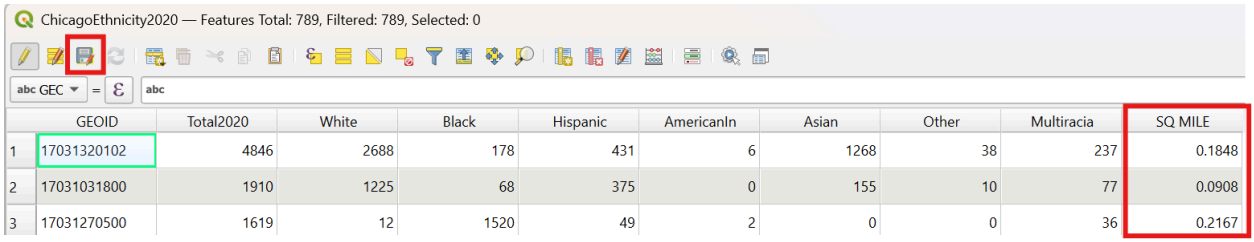


- In the Field calculator window, make the following changes:
  - Check the box next to **Create a new field** (we will create a new field to hold the area values)
  - Set the Output field name as **SQ MILE** (this is the name of the new field)

- Set the Output field type as **Decimal number (real)** (the data type of the new field)
- Set Output field width as **10 (total number of digits)**, **Precision** as **4** (**number of decimal places**)
- Enter the Expression to calculate **$area**. You may double-click the $area function from the **Geometry** section to add it.
- Click **OK** to calculate the polygon area in square miles.



- Be sure to save your changes by clicking on the **Save Edits** button  .



	GEOID	Total2020	White	Black	Hispanic	AmericanIn	Asian	Other	Multiracia	SQ MILE
1	17031320102	4846	2688	178	431	6	1268	38	237	0.1848
2	17031031800	1910	1225	68	375	0	155	10	77	0.0908
3	17031270500	1619	12	1520	49	2	0	0	36	0.2167

- Then click on the **Toggle editing mode** button  again to exit the editing mode. Then close the attribute table.